# Deep Spatio-Temporal Reasoning for Resilient Drone Swarm Formation in Dynamic Environments

***Abstract*—Orchestrating multi-agent systems in highly dynamic, partially observable environments necessitates a control paradigm capable of simultaneous spatial topology maintenance and temporal intent inference, yet existing approaches often decouple spatial aggregation from temporal reasoning, limiting their efficacy in scenarios demanding precise coordination under complex constraints. To bridge this gap, we propose a novel deep spatio-temporal reasoning framework tailored for multi-agent aerial coordination that unifies ego-centric graph attention, recurrent belief state modeling, and task-context awareness into an end-to-end differentiable policy. Distinct from static rule-based logic, our architecture operates under a Perceive, Predict, and Cooperate paradigm, employing a task-context attention mechanism to dynamically modulate the prioritization among diverse, potentially conflicting objectives based on the evolving situational context. To validate the efficacy and robustness of our approach, we instantiate the framework on a high-fidelity benchmark involving drone swarms navigating through non-linearly oscillating narrow gates task that imposes severe conflicts between geometric rigidity and trajectory compliance. Without manual curriculum design, the proposed method autonomously discovers an elastic formation strategy, where agents dynamically compress their topology to traverse bottlenecks and recover geometry post-traversal. Extensive empirical evaluations demonstrate that our framework significantly outperforms state-of-the-art optimization-based and learning-based baselines, achieving near-perfect success rates with superior inference efficiency and robust generalization across varying degrees of environmental dynamism.**

***Index Terms*—multi-agent reinforcement learning, spatio-temporal reasoning, graph neural networks, cooperative control.**

## I. Introduction

The transition of multi-agent systems (MAS) from controlled laboratory settings to complex, real-world environments has catalyzed a paradigm shift in autonomous control research [1]–[3]. In mission-critical applications—such as reliable formation flight in cluttered domains [4], [5], cooperative search and rescue in disaster zones [6], and large-scale environmental monitoring [7]—drone swarms are no longer afforded the luxury of operating in static, fully observable spaces. Instead, they must navigate environments characterized by strict geometric constraints and highly dynamic, non-linear threats. This operational reality imposes a severe dual challenge: agents must exhibit *spatial reasoning* to maintain safe, cohesive topologies for mission integrity, while simultaneously performing *temporal prediction* to anticipate and react to the non-linear evolution of dynamic obstacles [8]. The core difficulty lies in the potential conflict between these objectives: maintaining a rigid formation often contradicts the agility required to traverse narrow, time-varying bottlenecks, necessitating a control architecture capable of fluidly trading off structural rigidity for navigational compliance.

Despite the proliferation of MARL algorithms [9], [10], bridging the gap between high-level discrete coordination and low-level continuous control remains a formidable challenge, particularly in scenarios involving aggressive maneuvers. Classical approaches, specifically distributed model predictive control (DMPC) [11], [12] and optimization-based trajectory planning [4], [13], have long been the gold standard, providing theoretical guarantees for stability and collision avoidance via convex relaxation or linearization. However, these methods fundamentally suffer from the "horizon-latency" trade-off. To account for fast-moving, non-linear obstacles (such as oscillating gates), the prediction horizon must be extended, which causes the complexity of the optimization problem to scale cubically or worse. This results in prohibitive computational delays—often exceeding the timescale of the environment's dynamics—rendering the calculated trajectories obsolete by the time they are executed [14], [15]. Furthermore, optimization-based methods typically struggle with the non-convexity of formation reconfiguration, often getting trapped in conservative local optima where the swarm freezes rather than attempting a risky traversal.

Conversely, recent data-driven approaches, such as end-to-end learning for collision avoidance [16], [17] and multi-task control [18], offer agile, low-latency execution. However, standard MARL formulations often treat the multi-agent state as a flat vector or a simple permutation-invariant set, lacking the explicit structural priors necessary to maintain complex formation topologies. While recent advances have integrated graph neural networks (GNNs) [19], [20] to capture spatial interactions, and Transformers [21] to model long-term dependencies, a critical gap remains: most existing frameworks treat spatial aggregation and temporal recurrence as loosely coupled or independent modules [22], [23]. They rarely model the intricate *spatiotemporal coupling* between evolving neighborhood topologies and latent environmental physics (e.g., the unknown phase and frequency of an oscillating barrier). Moreover, in multi-objective tasks, standard scalarization techniques fail to dynamically adapt to changing priorities, leading to policies that are either too rigid to pass obstacles or too loose to maintain formation.

### *Contribution*

To bridge this gap, we propose graph attention recurrent task-aware MAPPO (GART-MAPPO), an end-to-end framework that unifies spatial perception, temporal prediction, and

task-aware decision-making under a principled "Perceive, Predict, and Cooperate" paradigm. Distinct from recent single-model multi-objective approaches [24], our architecture employs a hierarchical encoder that seamlessly integrates three specialized components: (1) ego-centric graph attention for scalable, permutation-invariant spatial reasoning; (2) gated recurrent units (GRUs) for efficient inference of latent environmental dynamics under partial observability; and (3) a novel task-context Attention mechanism. The latter is crucial for the "elasticity" of the system: it allows agents to dynamically modulate the attention weights of different task tokens (e.g., Formation Holding vs. Gate Traversal) based on real-time situational contexts.

We instantiate this framework on a high-fidelity benchmark: formation flight through non-linearly oscillating narrow gates. This task serves as a stress test for spatio-temporal reasoning, imposing a contradictory "elastic" constraint where agents must maintain rigid geometry for the majority of the flight but simultaneously deform to traverse bottlenecks that are narrower than the formation span [13], [25]. Without manual curriculum design or rule-based switching, GART-MAPPO autonomously discovers an "elastic formation" strategy—dynamically compressing to traverse narrow passages and recovering geometry post-traversal—effectively solving the spatio-temporal conflict that remains intractable for traditional rigid-body control.

The main contributions of this article are summarized as follows:

1) **Unified Spatio-Temporal Framework:** We establish a deep reasoning architecture that seamlessly couples graph-based spatial attention with recurrent latent representations. This design enables agents to infer unobservable system dynamics (e.g., target oscillation frequency) from historical interactions, achieving robust coordination in partially observable environments.
2) **Emergent Elastic Behavior:** We demonstrate the emergence of a sophisticated "elastic formation" strategy. By utilizing the proposed task-context attention mechanism, the framework balances conflicting objectives—formation rigidity versus geometric adaptation—autonomously transitioning between rigid flight and fluid deformation without heuristic state machines.
3) **Superior Performance and Efficiency:** Extensive empirical evaluations against adapted state-of-the-art baselines—representing both policy-search based MPC [15] and distributed trajectory optimization [5]—confirm that our approach achieves near-perfect success rates (97.91%). Crucially, our method maintains inference latencies ($\approx$1.5ms) orders of magnitude lower than model-based approaches, verifying its suitability for real-time deployment on agile aerial platforms.

## II. Related Work

### A. Optimization-Based Multi-Agent Control

The dominant paradigm for safe multi-agent coordination traditionally relies on explicit trajectory optimization and control theory. To ensure smoothness in trajectory generation, minimum-snap approaches [26] were early adopters, formulating the flight path as a polynomial optimization problem. Subsequently, DMPC [11] and consensus-based formation control [12] became standard techniques for ensuring inter-agent collision avoidance by solving constrained optimization problems at each time step. More recently, differentiable graph-based metrics have been introduced [4], [5] to quantify formation similarity, enabling robust trajectory planning that preserves global topology within dense environments. To further enhance safety guarantees, Control barrier functions (CBFs) have been integrated into distributed frameworks [27], acting as safety filters to modify nominal controls. In the specific domain of traversing narrow passages, affine deformable guidance strategies [28] have been proposed to mathematically model the formation as a deformable body, calculating feasible shapes to fit through constraints. However, these methods confine the swarm to a restricted set of geometric transformations (e.g., uniform scaling), limiting their flexibility in navigating irregular or highly non-convex narrow passages. While optimization-based methodologies provide theoretical safety assurances, they heavily depend on the accuracy of linearization models and global state availability [29]. In highly dynamic settings, such as the oscillating gate scenario in this work, the non-linear dynamics of the target violate the linear assumptions of standard MPC, while nonlinear MPC incurs prohibitive computational costs that degrade real-time responsiveness in agile scenarios. Furthermore, the computational latency inherent in online solvers creates a critical bottleneck: as the environment becomes more dynamic, the solver requires more iterations to converge, leading to outdated control actions [14], [15]. Our proposed learning-based framework circumvents this by shifting the computational burden to offline training, allowing for millisecond-level inference that is robust to non-linear dynamics and perceptual uncertainty.

### B. Spatio-Temporal Reasoning in MARL

To transcend the limitations of model-based control, deep reinforcement learning (DRL) has been extensively applied to high-dimensional continuous control for drone swarms [30]. Early end-to-end approaches utilizing attention mechanisms [17] demonstrated effective swarm collision avoidance with high zero-shot transferability but often lacked coordination for complex objectives. Addressing the unstructured nature of agent interactions, GNNs have emerged as a foundational tool. Seminal works [19] [20] leverage GNNs to model the communication topology, enabling scalable flocking and formation control. Parallel to this, Transformer-based architectures, including MAT [31] and its variants [21], have set new benchmarks by modeling the multi-agent problem as a sequence modeling task, demonstrating superior long-horizon reasoning. Other studies have explored heterogeneous graph structures [32] to accommodate diverse agent capabilities. Despite these advancements, a significant limitation persists in how these architectures handle temporal dynamics. Many

GNN-based methods [33] rely on stacking frame histories (e.g., Frame Stacking) to handle partial observability. This approach creates high-dimensional inputs and is often less efficient than explicit recurrent modeling for inferring latent physics (e.g., velocity, acceleration, external forces) [8]. Conversely, full Transformer architectures, while powerful, incur computational costs that grow quadratically with sequence length, precluding deployment on resource-constrained onboard hardware. Our GART-MAPPO framework strikes a strategic balance: it combines the spatial selectivity of graph attention networks (GATs) with the inference efficiency of GRUs, providing a lightweight yet powerful solution for real-time spatio-temporal reasoning.

### C. Flexible Formation and Multi-Objective Adaptation

The development of "flexible" or "elastic" formation control represents an active research frontier, aiming to reconcile structural rigidity with environmental adaptability. Formation-aware exploration strategies [25] and adaptable swarm frameworks [7] have been utilized to learn variable topologies that adjust to obstacle density. Similarly, distributed approaches for flexible coverage [34] and global-local balancing mechanisms [35] have demonstrated promising results in large-scale coordination. Fundamentally, achieving such flexibility requires solving a multi-objective optimization problem: balancing the conflicting goals of formation maintenance and navigational agility. In this domain, approaches like [24] typically employ single DRL models to optimize a static weighted sum of objectives. Furthermore, multi-task RL frameworks leveraging shared dynamics [18] have been introduced to transfer knowledge across diverse maneuvers. However, most existing flexible formation methods rely on pre-defined heuristic rules or manual curriculum designs to switch between rigid and loose modes. They often lack a unified mechanism to learn *when* to deform and *how much* to deform based on evolving contexts. Our work advances this domain by incorporating a learnable task-context attention module. While inspired by dual-objective attention mechanisms [36], our approach extends this concept to the temporal domain, enabling dynamic modulation of task tokens (e.g., gate relative position vs. neighbor relative position) in real-time. This allows the policy to autonomously discover the "elastic" strategy—prioritizing formation shape in open space and prioritizing traversal in constricted space—without requiring complex, manually tuned state machines.

## III. GART-MAPPO: A General Spatio-Temporal Cooperative Framework

This section details our GART-MAPPO (Graph-Attention Recurrent Task-aware MAPPO), an end-to-end framework designed to solve the "Perceive, Predict, and Cooperate" paradigm in complex multi-agent systems. As depicted in Fig. 1, the architecture orchestrates a hierarchical differentiable pipeline that unifies spatial perception, temporal belief modeling, and task-context awareness. Operating under the centralized training with decentralized execution (CTDE) paradigm, the framework ensures scalability during deployment while leveraging global information for stable policy optimization.

### A. Framework Overview and Notation

We consider a cooperative multi-agent system with $N$ homogeneous agents. While agents can perceive the relative states of their peers, the environment is partially observable regarding latent system dynamics (e.g., the unknown oscillation parameters of the gates). At each time step $t$, agent $i$ receives a structured local observation $\mathbf{o}_{i,t} \in \mathcal{O}$ and selects an action $\mathbf{a}_{i,t} \in \mathcal{A}$ according to its policy $\pi_{\theta_i}$. The observation space is explicitly decomposed into three semantic categories

$$\mathbf{o}_{i,t} = \left\{ \mathbf{o}_{i,t}^{self} \in \mathbb{R}^{d_s},\ \mathbf{O}_{i,t}^{others} \in \mathbb{R}^{(N-1)\times d_r},\ \mathbf{T}_{i,t} \in \mathbb{R}^{M\times d_\tau} \right\}, \tag{1}$$

where $\mathbf{o}_{i,t}^{self}$ denotes the agent's proprioceptive state (e.g., local position, linear velocity, and orientation), $\mathbf{O}_{i,t}^{others} = [\mathbf{o}_{i,1}^{rel}, \dots, \mathbf{o}_{i,N-1}^{rel}]^\top$ contains the relative states of all other agents, and $\mathbf{T}_{i,t} = [\boldsymbol{\tau}_1, \dots, \boldsymbol{\tau}_M]^\top$ represents $M$ task-specific tokens encoding environmental entities (e.g., dynamic targets, formation references, goal positions).

The policy is conditioned on both the current observation and a recurrent hidden state $\mathbf{h}_{i,t-1} \in \mathbb{R}^{d_h}$ that encodes temporal context

$$\pi_\theta(\mathbf{a}_{i,t}|\mathbf{o}_{i,t}, \mathbf{h}_{i,t-1}) = \mathcal{N}\left(\boldsymbol{\mu}_\theta(\mathbf{z}_{i,t}),\ \boldsymbol{\Sigma}_\theta\right), \tag{2}$$

where $\mathbf{z}_{i,t} = f_\theta(\mathbf{o}_{i,t}, \mathbf{h}_{i,t-1})$ is the latent representation produced by the encoder network $f_\theta$, and $\boldsymbol{\Sigma}_\theta$ is the covariance matrix. This modular observation structure enables instantiation across diverse multi-agent domains by simply configuring the content and dimensionality of task tokens.

### B. Spatial Perception: Ego-Centric Graph Attention

Traditional GNNs typically operate on fixed adjacency structures. To accommodate dynamic agent configurations and enable selective attention to task-relevant neighbors, we design an *ego-centric graph attention* mechanism. This module treats agent $i$ as the central node and constructs a fully connected local attention graph.

The input features are first projected into a common embedding space $\mathbb{R}^d$ via learnable linear transformations

$$\mathbf{e}_i^{ego} = \mathbf{W}_{self}\mathbf{o}_{i,t}^{self} \in \mathbb{R}^d, \quad \mathbf{E}_i^{others} = \mathbf{W}_{rel}\mathbf{O}_{i,t}^{others} \in \mathbb{R}^{(N-1)\times d}, \tag{3}$$

where $\mathbf{W}_{self} \in \mathbb{R}^{d\times d_s}$ and $\mathbf{W}_{rel} \in \mathbb{R}^{d\times d_r}$ are projection matrices. We construct the local node sequence by concatenating the ego embedding with neighbor embeddings

$$\mathbf{X}_i^{(0)} = \left[\mathbf{e}_i^{ego};\ \mathbf{E}_i^{others}\right] \in \mathbb{R}^{N\times d}. \tag{4}$$

This sequence is processed through $L$ stacked transformer encoder layers. Each layer $l$ comprises a Multi-Head Self-Attention (MHSA) mechanism followed by a Position-wise Feed-Forward Network (FFN). First, we compute the query, key, and value matrices

$$\mathbf{Q}^{(l)}, \mathbf{K}^{(l)}, \mathbf{V}^{(l)} = \mathbf{X}_i^{(l-1)}\mathbf{W}_Q^{(l)},\ \mathbf{X}_i^{(l-1)}\mathbf{W}_K^{(l)},\ \mathbf{X}_i^{(l-1)}\mathbf{W}_V^{(l)}, \tag{5}$$

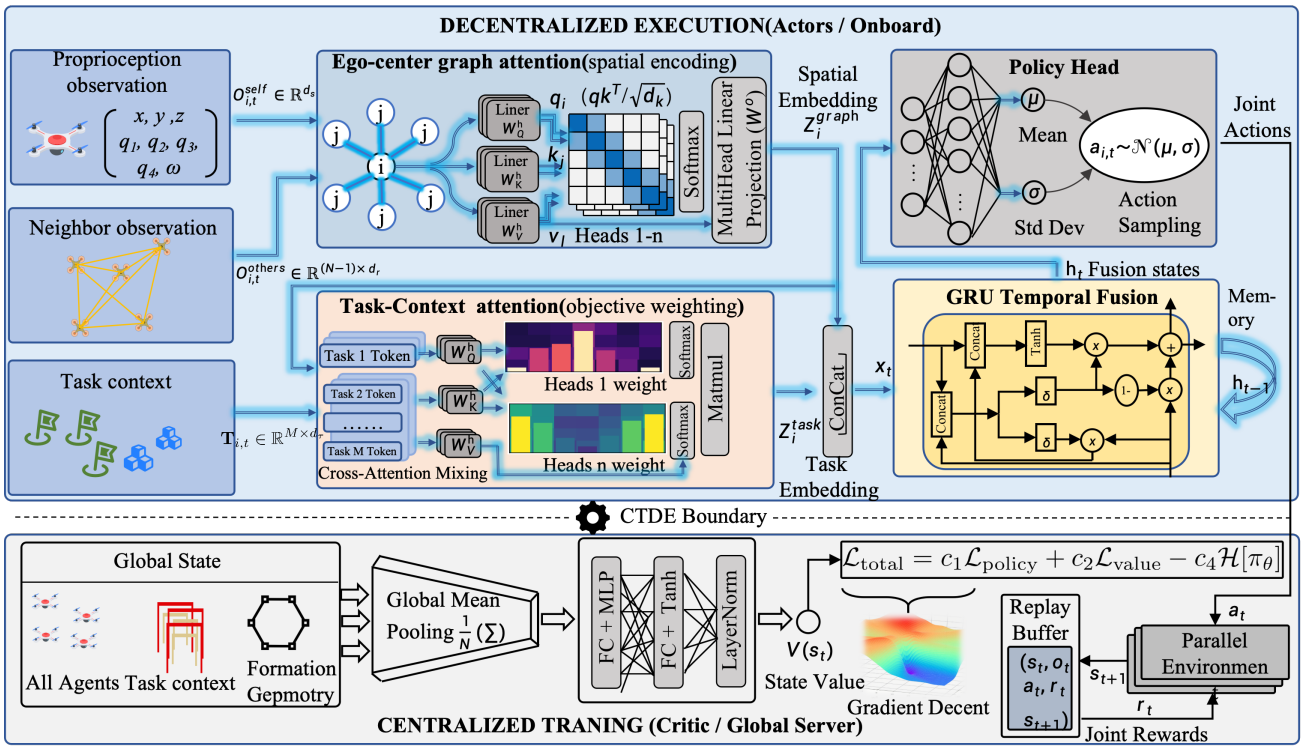



Fig. 1: Architectural overview of GART-MAPPO. The decentralized policy processes each agent's structured observation $\mathbf{o}_i$ through a hierarchical encoder: (1) **Ego-Centric Graph Attention** aggregates spatial neighborhood features; (2) **Task-Context Attention** dynamically weights conflicting objectives via cross-attention; and (3) **GRU Temporal Fusion** maintains a recurrent belief state $\mathbf{h}_t$ to infer latent dynamics. The resulting embedding drives a stochastic actor. During training, a centralized critic leverages global state information to guide policy updates. Dashed lines denote data flow exclusive to the centralized training phase.

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d}$. The attention weights are obtained via softmax

$$\mathbf{A}^{(l)} = \text{softmax}\left(\frac{\mathbf{Q}^{(l)}\mathbf{K}^{(l)\top}}{\sqrt{d/H}}\right), \tag{6}$$

where $H$ denotes the number of attention heads. The intermediate representation is computed via residual connection and layer normalization

$$\hat{\mathbf{X}}_i^{(l)} = \text{LayerNorm}\left(\mathbf{X}_i^{(l-1)} + \text{MHSA}(\mathbf{X}_i^{(l-1)})\right). \tag{7}$$

Subsequently, the sequence passes through the FFN

$$\mathbf{X}_i^{(l)} = \text{LayerNorm}\left(\hat{\mathbf{X}}_i^{(l)} + \text{FFN}(\hat{\mathbf{X}}_i^{(l)})\right). \tag{8}$$

The final spatial embedding $\mathbf{z}_i^{graph} \in \mathbb{R}^d$ is extracted from the ego-node position (index 0) of the output sequence $\mathbf{X}_i^{(L)}$, ensuring that the representation is focused on the decision-making agent

$$\mathbf{z}_i^{graph} = \mathbf{X}_i^{(L)}[0,:] \in \mathbb{R}^d. \tag{9}$$

**Remark 1** (Permutation Invariance)**.** *The ego-centric graph attention mechanism is permutation invariant with respect to the ordering of neighbor agents. For any permutation matrix* $\mathbf{P}$ *applied to the neighbor features, the resulting ego embedding remains unchanged:* $f_{graph}(\mathbf{o}^{self}, \mathbf{P}\mathbf{O}^{others}) = f_{graph}(\mathbf{o}^{self}, \mathbf{O}^{others})$*. This property ensures consistent policy behavior regardless of agent indexing conventions.*

### C. Objective Modulation: Task-Context Attention Module

Multi-agent tasks typically involve multiple, potentially conflicting objectives that require dynamic prioritization based on the current situational context. We introduce a *task-context attention* module that employs a multi-head cross-attention mechanism to compute task-aware embeddings.

Each task token $\boldsymbol{\tau}_m \in \mathbb{R}^{d_\tau}$ (representing entities such as dynamic gates, formation targets, or goal positions) is first projected into a canonical feature space

$$\mathbf{t}_m = \text{LayerNorm}\left(\mathbf{W}_{\tau_m}\boldsymbol{\tau}_m\right) \in \mathbb{R}^d, \quad m \in \{1,\dots,M\}, \tag{10}$$

where $\mathbf{W}_{\tau_m}$ are task-specific projection matrices. The task token matrix is formed as $\mathbf{T} = [\mathbf{t}_1;\dots;\mathbf{t}_M]^\top \in \mathbb{R}^{M\times d}$.

We utilize the agent's spatial embedding $\mathbf{z}_i^{graph}$ as the **Query**, seeking relevant guidance from the task tokens which serve as **Keys** and **Values**. Specifically, for each attention head $h$, we compute

$$\mathbf{q}^{(h)} = \mathbf{z}_i^{graph}\mathbf{W}_Q^{(h)}, \quad \mathbf{K}^{(h)} = \mathbf{T}\mathbf{W}_K^{(h)}, \quad \mathbf{V}^{(h)} = \mathbf{T}\mathbf{W}_V^{(h)}, \tag{11}$$

where $\mathbf{W}_Q^{(h)}, \mathbf{W}_K^{(h)}, \mathbf{W}_V^{(h)} \in \mathbb{R}^{d\times d_k}$ are learnable projection parameters.

The attention weights $\boldsymbol{\alpha}^{(h)} \in \mathbb{R}^M$ capture the relevance of each task objective given the current spatial configuration

$$\boldsymbol{\alpha}^{(h)} = \text{softmax}\left(\frac{\mathbf{q}^{(h)}\mathbf{K}^{(h)\top}}{\sqrt{d_k}}\right). \tag{12}$$

The task-aware context is then aggregated via a weighted sum and passed through a linear projection $\mathbf{W}_O$ to fuse multi-head outputs

$$\mathbf{z}_i^{task} = \left(\text{Concat}_{h=1}^{H}\left[\sum_{m=1}^{M}\alpha_m^{(h)}\mathbf{v}_m^{(h)}\right]\right)\mathbf{W}_O. \tag{13}$$

Crucially, the attention weights $\boldsymbol{\alpha}$ provide intuitive insights into the decision-making process, revealing explicitly whether the agent is prioritizing formation maintenance or obstacle traversal at any given moment.

### D. Temporal Inference: GRU-based Belief State

In partially observable environments, the true system state is not directly accessible. Agents must infer latent variables—such as the unobservable oscillation phase of dynamic gates or the intent of neighbors—from the history of observations. We employ a Gated Recurrent Unit (GRU) to maintain a *belief state* that encodes temporal context efficiently.

First, the spatial embedding $\mathbf{z}_i^{graph}$ and task-aware embedding $\mathbf{z}_i^{task}$ are fused into a unified observation representation via a non-linear projection:

$$\mathbf{f}_{i,t} = \text{GELU}\left(\text{LayerNorm}\left(\mathbf{W}_{fuse}\left[\mathbf{z}_i^{graph}\|\mathbf{z}_i^{task}\right]\right)\right) \in \mathbb{R}^{d_f}, \tag{14}$$

where $\|$ denotes concatenation and $\mathbf{W}_{fuse} \in \mathbb{R}^{d_f \times 2d}$ is the learnable weight matrix.

This fused feature vector serves as the input to the recurrent unit. We utilize a GRU due to its effectiveness in mitigating the vanishing gradient problem while being computationally lighter than LSTM:

$$\mathbf{h}_{i,t} = \text{GRU}(\mathbf{f}_{i,t}, \mathbf{h}_{i,t-1}; \boldsymbol{\theta}_{gru}), \tag{15}$$

where $\mathbf{h}_{i,t} \in \mathbb{R}^{d_h}$ represents the recurrent belief state update based on the current input and previous history. Note that $\mathbf{h}_{i,t}$ is reset to zero at the beginning of each episode to prevent cross-episode interference.

To ensure the policy remains responsive to immediate sensory stimuli while leveraging historical context, we employ a skip connection for the final representation. The current fused feature $\mathbf{f}_{i,t}$ is concatenated with the updated history $\mathbf{h}_{i,t}$ before passing through the final decoding layers:

$$\mathbf{z}_{i,t} = \text{MLP}_{post}\left(\text{LayerNorm}\left([\mathbf{f}_{i,t}\|\mathbf{h}_{i,t}]\right)\right) \in \mathbb{R}^{d_{out}}. \tag{16}$$

This latent representation $\mathbf{z}_{i,t}$ effectively decouples the "what is happening now" (via $\mathbf{f}_{i,t}$) from "what is the latent trend" (via $\mathbf{h}_{i,t}$), providing a robust state estimate for the subsequent actor-critic heads.

### E. Policy Optimization via Recurrent MAPPO

We adopt the Recurrent MAPPO algorithm [9], which extends the standard PPO framework to multi-agent settings under the CTDE paradigm. This section details the actor-critic heads and the truncated backpropagation through time (TBPTT) optimization strategy.

The actor network maps the latent representation $\mathbf{z}_{i,t}$ (from Eq. 16) to a stochastic policy. Following the maximum entropy paradigm, we parameterize the action distribution as a diagonal Gaussian

$$\pi_\theta(\mathbf{a}_{i,t}|\mathbf{z}_{i,t}) = \mathcal{N}\left(\boldsymbol{\mu}_\theta(\mathbf{z}_{i,t}), \text{diag}(\boldsymbol{\sigma}^2)\right), \tag{17}$$

where the mean $\boldsymbol{\mu}_\theta$ is generated by a two-layer MLP with Mish activation, followed by a Tanh squashing function to bound actions within feasible control limits $[-1, 1]$. For exploration stability, the standard deviation $\boldsymbol{\sigma}$ is parameterized as a learnable, state-independent vector.

During centralized training, the critic network approximates the global value function $V_\phi(\mathbf{s}_t^{central})$ to reduce variance in gradient estimation. The centralized state $\mathbf{s}_t^{central}$ aggregates information from the entire swarm and the environment. To ensure scalability and permutation invariance regarding the swarm size, the critic utilizes a global pooling architecture

$$V_\phi(\mathbf{s}_t^{central}) = \text{MLP}_{critic}(\text{Concat}[\bar{\mathbf{z}}^{agents}\|\bar{\mathbf{z}}^{targets}\| \bar{\mathbf{z}}^{formation}]), \tag{18}$$

where $\bar{\mathbf{z}}^{(\cdot)}$ denotes the mean-pooled embeddings of agent features, task tokens, and formation references, respectively.

The parameters of the encoder, actor, and critic are optimized jointly. To effectively train the recurrent components (GRU) over long-horizon episodes without suffering from vanishing gradients or excessive memory consumption, we employ truncated backpropagation through time (TBPTT). The episode is divided into sequences of length $T_{chunk}$. The generalized advantage estimation is computed within each chunk, and gradients are backpropagated only through the current sequence, treating the initial hidden state as a constant. The joint objective function is defined as

$$L(\theta, \phi) = \mathbb{E}_t\left[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\phi) + c_2 S[\pi_\theta](\mathbf{z}_{i,t})\right], \tag{19}$$

where $L_t^{CLIP}(\theta)$ is the standard PPO clipped surrogate objective to constrain policy updates, $L_t^{VF}(\phi)$ is the value function squared-error loss, and $S[\pi_\theta]$ denotes the entropy bonus to encourage exploration. $c_1$ and $c_2$ are coefficients balancing value accuracy and exploration. The complete training procedure of GART-MAPPO is summarized in Algorithm 1.

## IV. Validation Scenario: Formation Flight Through Dynamic Gates

To rigorously evaluate the spatio-temporal reasoning capabilities of GART-MAPPO, we instantiate the framework within a high-fidelity benchmark environment: a drone swarm navigating through non-linearly oscillating narrow gates. This section delineates the physical modeling, the mathematical formulation of the underlying Markov Decision Process (MDP), and the specific instantiation details.

### A. Physical System Modeling

We begin by formalizing the agent dynamics. Consider a swarm of $N$ homogeneous quadrotors operating in three-

**Algorithm 1** GART-MAPPO Training with TBPTT

**Initialize:** Actor $\theta$, Critic $\phi$, Value Normalizer $\mathcal{V}$
**Hyperparameters:** $T_{roll}$, $L_{seq}$, $K_{epochs}$, $\epsilon$, $\gamma$, $\lambda$
**for** iteration $= 1, 2, \ldots$ **do**
  Collect trajectories $\mathcal{D} = \{(\mathbf{o}_t, \mathbf{a}_t, r_t, \mathbf{h}_t)\}_{t=1}^{T_{roll}}$ using $\pi_{\theta_{old}}$
  Compute GAE advantages $\{\hat{A}_t\}$ and returns $\{\hat{R}_t\}$
  Normalize advantages: $\hat{A}_t \leftarrow (\hat{A}_t - \mu_A)/\sigma_A$
  Normalize value targets: $\hat{R}_t \leftarrow \mathcal{V}.\text{normalize}(\hat{R}_t)$
  **for** epoch $= 1, \ldots, K_{epochs}$ **do**
    Segment $\mathcal{D}$ into sequences of length $L_{seq}$
    **for** each minibatch $\mathcal{B}$ **do**
      Recompute $\{\log \pi_\theta, \mathcal{H}, V_\phi\}$ with TBPTT over $\mathcal{B}$
      Update $\theta \leftarrow \theta - \alpha_\theta \nabla_\theta \mathcal{L}^{actor}$
      Update $\phi \leftarrow \phi - \alpha_\phi \nabla_\phi \mathcal{L}^{critic}$
    **end for**
  **end for**
  Update Value Normalizer: $\mathcal{V}.\text{update}(\{\hat{R}_t\})$
**end for**

dimensional space. The state of the $i$-th agent is defined as $\mathbf{s}_i = [\mathbf{p}_i, \mathbf{v}_i, \mathbf{q}_i]^\top$, where $\mathbf{p}_i \in \mathbb{R}^3$ and $\mathbf{v}_i \in \mathbb{R}^3$ represent the position and linear velocity in the world frame, respectively. The attitude is represented by a unit quaternion $\mathbf{q}_i \in \mathbb{H}$.

The control input for the $i$-th agent is defined as $\mathbf{u}_i = [f_i, \boldsymbol{\omega}_i]^\top$, where $f_i \in \mathbb{R}_{\geq 0}$ is the collective thrust magnitude and $\boldsymbol{\omega}_i \in \mathbb{R}^3$ is the body angular velocity vector.

The translational dynamics consist of the position kinematics

$$\dot{\mathbf{p}}_i = \mathbf{v}_i, \tag{20}$$

and the velocity dynamics governed by gravity and thrust

$$\dot{\mathbf{v}}_i = \mathbf{g} + \frac{f_i}{m}\mathbf{R}(\mathbf{q}_i)\mathbf{e}_3, \tag{21}$$

where $m$ is the mass of the quadrotor, $\mathbf{g} = [0, 0, -9.81]^\top$ is the gravity vector, $\mathbf{e}_3 = [0, 0, 1]^\top$ is the unit vector along the body $z$-axis, and $\mathbf{R}(\mathbf{q}_i)$ is the rotation matrix derived from the quaternion $\mathbf{q}_i$.

The rotational dynamics are described by

$$\dot{\mathbf{q}}_i = \frac{1}{2}\mathbf{q}_i \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega}_i \end{bmatrix}, \tag{22}$$

where $\otimes$ denotes the quaternion multiplication.

We discretize these continuous-time dynamics using the 4th-order Runge-Kutta (RK4) method

$$\mathbf{s}_{i,t+1} = \mathbf{s}_{i,t} + \frac{\Delta t}{6}(K_1 + 2K_2 + 2K_3 + K_4), \tag{23}$$

where $\Delta t$ is the simulation time step. The intermediate increments are calculated sequentially. The first increment evaluates the dynamics at the current state

$$K_1 = f_{dyn}(\mathbf{s}_{i,t}, \mathbf{u}_{i,t}). \tag{24}$$

The second and third increments use half-step predictions

$$K_2 = f_{dyn}\left(\mathbf{s}_{i,t} + \frac{\Delta t}{2}K_1, \mathbf{u}_{i,t}\right), \tag{25}$$

$$K_3 = f_{dyn}\left(\mathbf{s}_{i,t} + \frac{\Delta t}{2}K_2, \mathbf{u}_{i,t}\right). \tag{26}$$

The fourth increment uses a full-step prediction

$$K_4 = f_{dyn}(\mathbf{s}_{i,t} + \Delta t K_3, \mathbf{u}_{i,t}). \tag{27}$$

To rigorously quantify the formation topology and facilitate the "elastic" behaviors where the swarm must maintain shape fidelity under varying scales, we adopt a graph-theoretic formulation invariant to translation, rotation, and uniform scaling. We model the swarm's spatial topology as a complete undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the node set $\mathcal{V} = \{1, \ldots, N\}$ represents the individual agents.

Distinct from rigid formation definitions that rely on fixed Euclidean offsets, we characterize the formation geometry through the spectral properties of the interaction graph. We define the weighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, where the edge weights correspond to the squared Euclidean distance between agent pairs, capturing the global geometric constellation

$$A_{ij} = \begin{cases} \|\mathbf{p}_i - \mathbf{p}_j\|^2 & \text{if } i \neq j, \\ 0 & \text{otherwise.} \end{cases} \tag{28}$$

The structural information is encapsulated in the symmetric normalized Laplacian matrix $\hat{\mathbf{L}}$. First, we compute the degree matrix $\mathbf{D} = \text{diag}(d_1, \ldots, d_N)$, where each diagonal element $d_i = \sum_{j=1}^{N} A_{ij}$ represents the cumulative squared distance to all other agents. The normalized Laplacian is then derived as

$$\hat{\mathbf{L}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}. \tag{29}$$

A critical property of $\hat{\mathbf{L}}$ for our application is its invariance to the formation's scale and rigid body transformations. This allows us to define a desired formation shape $\hat{\mathbf{L}}_{\text{ref}}$ based on a nominal configuration and evaluate the swarm's topological error regardless of whether the formation is expanded or compressed. During the task, minimizing the Frobenius norm difference $\|\hat{\mathbf{L}} - \hat{\mathbf{L}}_{\text{ref}}\|_F$ ensures the agents maintain the triangular geometry, even as they dynamically contract inter-agent distances to negotiate narrow gates.

The dynamic target is modeled as a nonlinear pendulum, approximated as a point mass suspended on a weightless and inextensible string of length $L_{cm}$ from a fixed support at position $\boldsymbol{P}_{WP} = [x_f, y_f, z_f]$. The dynamics of this rotational motion are

$$\ddot{\theta}_x = -\left(\frac{m g_z L_{cm} \sin\theta_x}{I} + b\dot{\theta}_x\right), \quad \ddot{\theta}_y = 0, \quad \ddot{\theta}_z = 0 \tag{30}$$

where $\theta_x$ is the roll angle, $I$ is the moment of inertia, and $b \in \mathbb{R}_+$ is a damping factor. The translational motion is

$$\ddot{v}_x = 0, \quad \ddot{v}_y = L_{cm}\cos(\theta_x)\ddot{\theta}_x, \quad \ddot{v}_z = L_{cm}\sin(\theta_x)\ddot{\theta}_x \tag{31}$$

The full state of the gate center is represented as $\mathbf{x}^g = [\mathbf{p}^g, \mathbf{q}^g, \mathbf{v}^g]$.

### B. Problem Formulation

The dynamic gate traversal task is formalized as a finite-horizon constrained optimization problem. We consider a swarm of $N$ agents characterized by state trajectories $\mathcal{X} = \{\mathbf{x}_1(t), \ldots, \mathbf{x}_N(t)\}$ and control policies $\mathcal{U} = \{\mathbf{u}_1(t), \ldots, \mathbf{u}_N(t)\}$ over a time horizon $t \in [0, T]$. The overarching objective is to minimize a joint cost functional

$\mathcal{J}$, subject to rigid body dynamics, safety requirements, and the time-varying geometric constraints imposed by the environment.

The optimization objective is defined as the sum of terminal and running costs. The terminal cost penalizes deviations from the target endpoint $\mathbf{p}_{goal}$ with $\phi(\mathbf{x}_i(T)) = \|\mathbf{p}_i(T) - \mathbf{p}_{goal}\|^2$. The complete objective function is

$$\min_{\mathcal{U}} \mathcal{J} = \sum_{i=1}^{N} \Bigg( \phi(\mathbf{x}_i(T)) + \int_0^T \left(\mathcal{L}_{form}(\mathbf{x}(t)) + \mathcal{L}_{reg}(\mathbf{u}_i(t))\right) dt \Bigg). \tag{32}$$

where the running cost comprises two terms: $\mathcal{L}_{form}$, which quantifies the topological deviation from the reference formation geometry to ensure structural coherence; and $\mathcal{L}_{reg} = \|\mathbf{u}_i(t)\|^2$, which regularizes the control effort to facilitate smooth maneuvering.

The optimization is governed by a set of strict physical and operational constraints. First, the evolution of each agent's state must satisfy the quadrotor dynamics $\dot{\mathbf{x}}_i(t) = f_{dyn}(\mathbf{x}_i(t), \mathbf{u}_i(t))$ as detailed in Eq. (26)-(36). Second, to guarantee operational safety, the Euclidean distance between any pair of agents is strictly bounded by a safety radius $r_{safe}$, such that $\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| \geq r_{safe}$ for all $i \neq j$.

The distinct complexity of this mission arises from the *spatio-temporal bottleneck* constraint. Let $\mathcal{O}_{frame}(t) \subset \mathbb{R}^3$ denote the time-varying collision volume of the oscillating gate frame, and $\mathcal{A}_{aperture}(t) \subset \mathbb{R}^3$ represent the safe passage volume within the frame. The valid trajectory must satisfy

$$\mathbf{p}_i(t) \notin \mathcal{O}_{frame}(t), \quad \forall t \in [0, T], \tag{33}$$

while simultaneously ensuring that there exists a specific traversal timestamp $t_{pass} \in [0, T]$ for which $\mathbf{p}_i(t_{pass}) \in \mathcal{A}_{aperture}(t_{pass})$. This condition necessitates that agents actively avoid the dynamic structure while synchronizing their intersection with the aperture.

Furthermore, a critical geometric conflict exists: the aperture width $W_{gate}$ is strictly narrower than the nominal formation span $W_{form}$. Consequently, the swarm is forced to temporarily relax its rigid geometric constraints during the traversal phase, satisfying

$$\max_{i,j} \|\mathbf{p}_i(t_{pass}) - \mathbf{p}_j(t_{pass})\| < W_{gate}. \tag{34}$$

This requirement demands an "elastic" formation strategy. Solving this non-convex optimization problem directly via classical MPC proves computationally intractable due to the highly non-linear nature of the gate dynamics and the prohibitive horizon length required to predict the oscillation phase. Therefore, we employ the GART-MAPPO framework to approximate the optimal policy $\pi_\theta$ in an end-to-end manner.

### C. MDP Instantiation

We formulate this task as a Partially Observable Markov Decision Process (POMDP), defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle$.

The observation vector $\mathbf{o}_i$ for agent $i$ is structured as $\mathbf{o}_i = [\mathbf{s}_i^{ego}, \mathbf{s}_i^{neigh}, \mathbf{s}_i^{gate}, \mathbf{s}_i^{form}, \mathbf{s}_i^{end}]$, where each component serves a distinct perceptual role. The ego state $\mathbf{s}_i^{ego} = [\mathbf{p}_i, \mathbf{v}_i, \mathbf{q}_i, \boldsymbol{\omega}_i] \in \mathbb{R}^{13}$ encapsulates the agent's position ($\mathbf{p}_i \in \mathbb{R}^3$), linear velocity ($\mathbf{v}_i \in \mathbb{R}^3$), orientation quaternion ($\mathbf{q}_i \in \mathbb{R}^4$), and angular velocity ($\boldsymbol{\omega}_i \in \mathbb{R}^3$), all expressed in a shared environment-centered coordinate frame. The neighbor state $\mathbf{s}_i^{neigh} = \{[\mathbf{p}_j, \mathbf{v}_j, \mathbf{q}_j, \boldsymbol{\omega}_j]\}_{j \in \mathcal{N}_i} \in \mathbb{R}^{(N-1)\times 13}$ provides the full kinematic states of all other agents in the formation, where $N$ is the total number of agents. The gate observation $\mathbf{s}_i^{gate} = [\mathbf{c}_1^g, \mathbf{c}_2^g, \mathbf{c}_3^g, \mathbf{c}_4^g, \mathbf{v}^g, \boldsymbol{\omega}^g] \in \mathbb{R}^{18}$ encodes the current target gate's geometry and motion, where $\{\mathbf{c}_k^g\}_{k=1}^4$ represent the four corner positions of the gate (12 dimensions) in the environment-centered frame, $\mathbf{v}^g \in \mathbb{R}^3$ is the gate's linear velocity, and $\boldsymbol{\omega}^g \in \mathbb{R}^3$ is its angular velocity. This corner-based representation explicitly captures the gate's orientation and size, enabling agents to plan traversal trajectories through the dynamic aperture. The formation target $\mathbf{s}_i^{form} = \mathbf{p}_i - \mathbf{p}_i^{target} \in \mathbb{R}^3$ represents the position error from the agent's designated formation position, where $\mathbf{p}_i^{target}$ is computed from the nominal formation configuration centered at the current formation centroid. This provides direct feedback for formation maintenance. The endpoint information $\mathbf{s}_i^{end} = [\mathbf{p}^{end} - \bar{\mathbf{p}}, d^{end}] \in \mathbb{R}^4$ encodes the navigation goal, where $\mathbf{p}^{end}$ is the centroid of the final endpoint positions, $\bar{\mathbf{p}}$ is the current formation center, and $d^{end} = \|\mathbf{p}^{end} - \bar{\mathbf{p}}\|$ is the distance to the endpoint. This representation enables coordinated progress toward the final destination. Crucially, agents do not observe latent physical parameters such as the pendulum length, damping coefficient, or initial gate angular states. The gate motion is governed by unobserved pendulum dynamics with parameters randomly sampled at episode initialization. This partial observability necessitates the temporal prediction module to infer motion patterns from observation history, enabling agents to anticipate future gate positions for successful traversal planning.

The action vector $\mathbf{a}_i \in [-1, 1]^4$ is defined as

$$\mathbf{a}_i = [f_{cmd}, \omega_{x,cmd}, \omega_{y,cmd}, \omega_{z,cmd}]^\top, \tag{35}$$

where $f_{cmd}$ represents the normalized collective thrust and $\omega_{\cdot,cmd}$ represents the normalized target body rates in the roll, pitch, and yaw axes. These high-level commands are tracked by an onboard flight controller, allowing the reinforcement learning policy to focus on trajectory generation rather than low-level motor control.

The reward function embodies the "elastic formation" concept by balancing formation maintenance against the necessity for temporary deformation. We formulate the total reward $R_t^i$ as the sum of three components

$$R_t^i = R_{form}^i + R_{task}^i + R_{reg}^i. \tag{36}$$

The formation integrity component employs a Laplacian-based formulation that is inherently scale-invariant. The formation shape reward uses a Cauchy kernel to provide smooth gradients

$$R_{shape} = \frac{1}{1 + \left(\|\mathbf{L} - \mathbf{L}_{ref}\|_F / (N \cdot \sigma_{form})\right)^2}, \tag{37}$$

where $\mathbf{L}_{ref}$ is the reference Laplacian computed from the nominal formation, $N$ is the number of agents, and $\sigma_{form}$ is

a tolerance parameter. The formation size reward constrains the maximum pairwise distance $d_{max} = \max_{i,j} \|\mathbf{p}_i - \mathbf{p}_j\|$ to match the nominal formation span $d_{ref}$

$$R_{size} = \frac{1}{1 + ((d_{max} - d_{ref})/\sigma_{size})^2}. \tag{38}$$

A soft separation penalty prevents inter-agent collision when the minimum distance $d_{min} = \min_{i \neq j} \|\mathbf{p}_i - \mathbf{p}_j\|$ falls below the safety threshold $d_{safe}$

$$P_{sep} = \max(0, d_{safe} - d_{min}). \tag{39}$$

To enable elastic deformation near the gate, the shape reward weight is modulated by the distance $d_{gate}$ from the formation center to the current gate: $\kappa_{elastic} = \text{clamp}(d_{gate}/d_{elastic}, 0.2, 1.0)$, where $d_{elastic}$ defines the region where formation constraints are relaxed. The complete formation reward is

$$R^i_{form} = w_{lap} \cdot \kappa_{elastic} \cdot R_{shape} + w_{size} \cdot R_{size} - w_{sep} \cdot P_{sep}. \tag{40}$$

The task progression component drives agents through the dynamic gate towards the final endpoint. The gate traversal reward is awarded per agent upon successful passage through the gate's geometric boundary

$$R^i_{task} = w_{trav} \cdot \mathbb{I}^i_{pass} + w_g \cdot (\mathbf{v}_i \cdot \hat{\mathbf{d}}^i_{gate}) \cdot \mathbb{I}^i_{pre} + w_e \cdot (\mathbf{v}_i \cdot \hat{\mathbf{d}}^i_{end}) \cdot \mathbb{I}^i_{post}, \tag{41}$$

where $\mathbb{I}^i_{pass}$ indicates successful gate traversal by agent $i$, $\hat{\mathbf{d}}^i_{gate}$ and $\hat{\mathbf{d}}^i_{end}$ are the unit direction vectors from agent $i$ to the gate and endpoint respectively, $\mathbb{I}^i_{pre}$ is active before gate passage to encourage approach, and $\mathbb{I}^i_{post}$ is active after passage to guide endpoint convergence. This stage-aware formulation prevents conflicting gradients and provides clear progression signals.

Finally, to guarantee the generation of physically feasible and smooth control policies, we incorporate a regularization component $R^i_{reg}$. This term explicitly suppresses high-frequency control oscillations and ensures optimal sensor orientation, defined as the weighted sum of three sub-objectives:

$$\begin{aligned} R^i_{reg} = & \underbrace{w_{thr} \cdot \text{clamp}(\tau_{thr} - |\Delta f_i|, 0)}_{R_{thr}\text{: Throttle Smoothness}} \\ & - \underbrace{w_{spin} \cdot |\omega_{z,i}|}_{R_{spin}\text{: Yaw Stability}} \\ & + \underbrace{w_{head} \cdot (\hat{\mathbf{x}}_{B,i} \cdot \hat{\mathbf{d}}^i_{target})}_{R_{head}\text{: Heading Alignment}}. \end{aligned} \tag{42}$$

Specifically, $R_{thr}$ penalizes abrupt thrust changes to prevent motor saturation and mechanical wear; $R_{spin}$ mitigates rotational instability by suppressing excessive yaw rates; and $R_{head}$ encourages the agent to align its body frame (and front-facing sensors) with the target direction, facilitating continuous visual tracking of the gate.

## V. Experiments and Analysis

To rigorously validate the efficacy of the GART-MAPPO framework, we conducted a comprehensive empirical evaluation centered on the formation flight task through dynamic gates. The experimental design is structured to systematically address four core research questions: the adaptability of the swarm to extreme geometric constraints via elastic mechanisms, the comparative performance against state-of-the-art optimization and learning-based paradigms, the policy robustness under varying task complexities, and the interpretability of the proposed attention-based architecture.

### A. Experimental Setup and Methodology

We instantiated the validation environment within the NVIDIA Isaac Sim platform, constructing a high-fidelity physics simulation to mirror real-world dynamics. The experimental subject comprises a swarm of $N = 3$ quadrotors, modeled with the physical parameters of the Bitcraze Crazyflie 2.1 ($m = 0.027$ kg). To rigorously test the spatial reasoning capabilities, the agents are initialized in a V-shaped formation defined by the relative coordinates $\{[0, 0, 0], [-0.3, 0.3, 0], [-0.3, -0.3, 0]\}$ (in meters). This topological configuration results in a nominal lateral formation span of $W_{form} \approx 0.6$ m. Crucially, the swarm is tasked with traversing a non-linearly oscillating gate characterized by a physical aperture width of $W_{gate} = 0.4$ m. The resulting geometric disparity, where $W_{gate} < W_{form}$, imposes a stringent constraint that necessitates autonomous, coordinated elastic contraction to negotiate the bottleneck without collision. The gate introduces significant temporal dynamism by oscillating with a variable period ranging from 2.5 s to 4.0 s, precluding the use of pre-computed static trajectories.

Policy optimization was conducted using the OmniDrones framework. To ensure robust convergence and explore the state space extensively, the training process spanned $5 \times 10^9$ aggregate simulation frames across 1,024 parallel environments. All computational experiments were executed on a workstation equipped with an Intel i9-13900KF CPU and an NVIDIA RTX 4080 GPU.

To benchmark the proposed approach against the spectrum of existing solutions, we selected three representative baselines. First, to represent hybrid planning methods, we employed an augmented version of *High-MPC* [15], which combines policy search with Model Predictive Control; we enhanced its original cost function with formation maintenance terms to support cooperative flight. Second, representing distributed trajectory optimization, we adapted *Dist-Opt* [5], a state-of-the-art method for robust formation maintenance. To accommodate the dynamic scenario, the real-time gate position was explicitly fed as a reference waypoint to its solver. Finally, *MLP-MAPPO* serves as an ablation baseline. This variant utilizes standard multi-layer perceptrons, excluding the proposed ego-centric graph attention and recurrent belief modules, thereby isolating the contribution of our specific spatio-temporal reasoning mechanisms.

Performance quantification relies on a multi-dimensional metric suite designed to capture different aspects of swarm behavior. The primary efficacy indicator is the *Success Rate (SR)*, defined as the percentage of episodes where all agents

TABLE I: Quantitative Performance Comparison under Randomized Gate Dynamics

| Method | Success Rate (%) | Inference Time (ms) | Laplacian Cost (↓) | Formation Size (m) | Min Sep. (m) | Velocity (m/s) | | Ang. Vel. (rad/s) | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Avg | Max | Avg | Max |
| High-MPC | 40.00 | 79.19 | 0.1594 | 0.6648 | 0.3351 | 2.34 | 4.30 | 2.38 | 10.16 |
| Dist-Opt | 10.00 | 2.48 | 0.4864 | 2.1717 | 0.9351 | 0.61 | 1.03 | 0.47 | 5.05 |
| **GART-MAPPO** | **97.91** | **1.51** | **0.1542** | 0.7073 | 0.4602 | 2.97 | 5.62 | 7.24 | 34.84 |

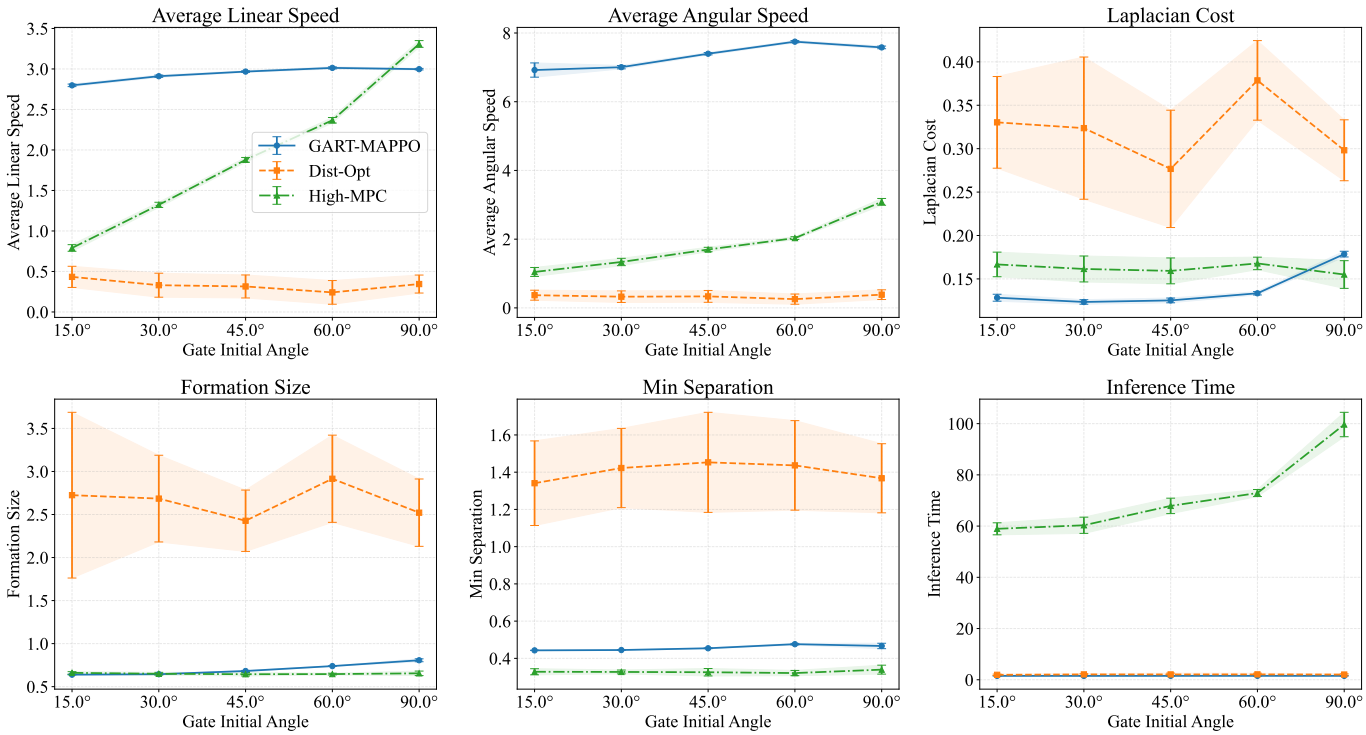



Fig. 2: Algorithm Performance Under Different Initial Swing Angles of Gate. The independent variable (Gate Initial Angle) acts as a proxy for task difficulty. **Top Row:** GART-MAPPO maintains high linear speed and exceptional angular agility while keeping formation error (Laplacian Cost) low. **Bottom Row:** High-MPC suffers from linearly increasing computational cost, while Dist-Opt fails to maintain formation compactness.

successfully traverse the gate without collision. Computational efficiency is evaluated via *Inference Time*, measuring the wall-clock time per policy step. Geometric fidelity is monitored through the *Formation Error* (Frobenius norm of the Laplacian difference) and *Minimum Separation*, which safeguards against inter-agent collisions. Additionally, *Flight Agility* metrics, including velocity and angular rate profiles, are analyzed to assess the smoothness and feasibility of the generated maneuvers.

### B. Comparative Performance Analysis

To rigorously assess the framework's capability under extreme uncertainty, we conducted stress tests in a highly dynamic scenario characterized by randomized static release angles. The quantitative outcomes are summarized in Table I.

As evidenced by the statistical aggregation in Table I, GART-MAPPO achieves a dominant success rate of 97.91%, significantly outperforming the baselines. A detailed inspection of the flight metrics reveals divergent failure modes for the contrasting control paradigms. The Dist-Opt baseline, despite its negligible inference latency (2.48 ms), exhibits a catastrophic failure in coordination. It records an exceptionally low average velocity (0.61 m/s) and, crucially, a massive deviation in formation size (2.17 m) and high Laplacian cost (0.4864). These metrics suggest that the gradient-based optimizer could not find a feasible solution that satisfied both the rigid formation constraints and the narrow gate aperture. Consequently, the swarm effectively disbanded—sacrificing topological integrity to avoid collisions—and hovered conservatively outside the critical interaction zone, leading to a near-zero success rate.

Conversely, High-MPC maintained a tight formation (0.66 m) and competitive velocity, yet achieved a success rate of only 40.00%. The primary bottleneck is identified as the "horizon-latency" trade-off. The average inference time of 79.19 ms introduces a critical delay between perception and actuation. In scenarios with rapid non-linear oscillations, this latency renders the planned trajectory obsolete by the time of

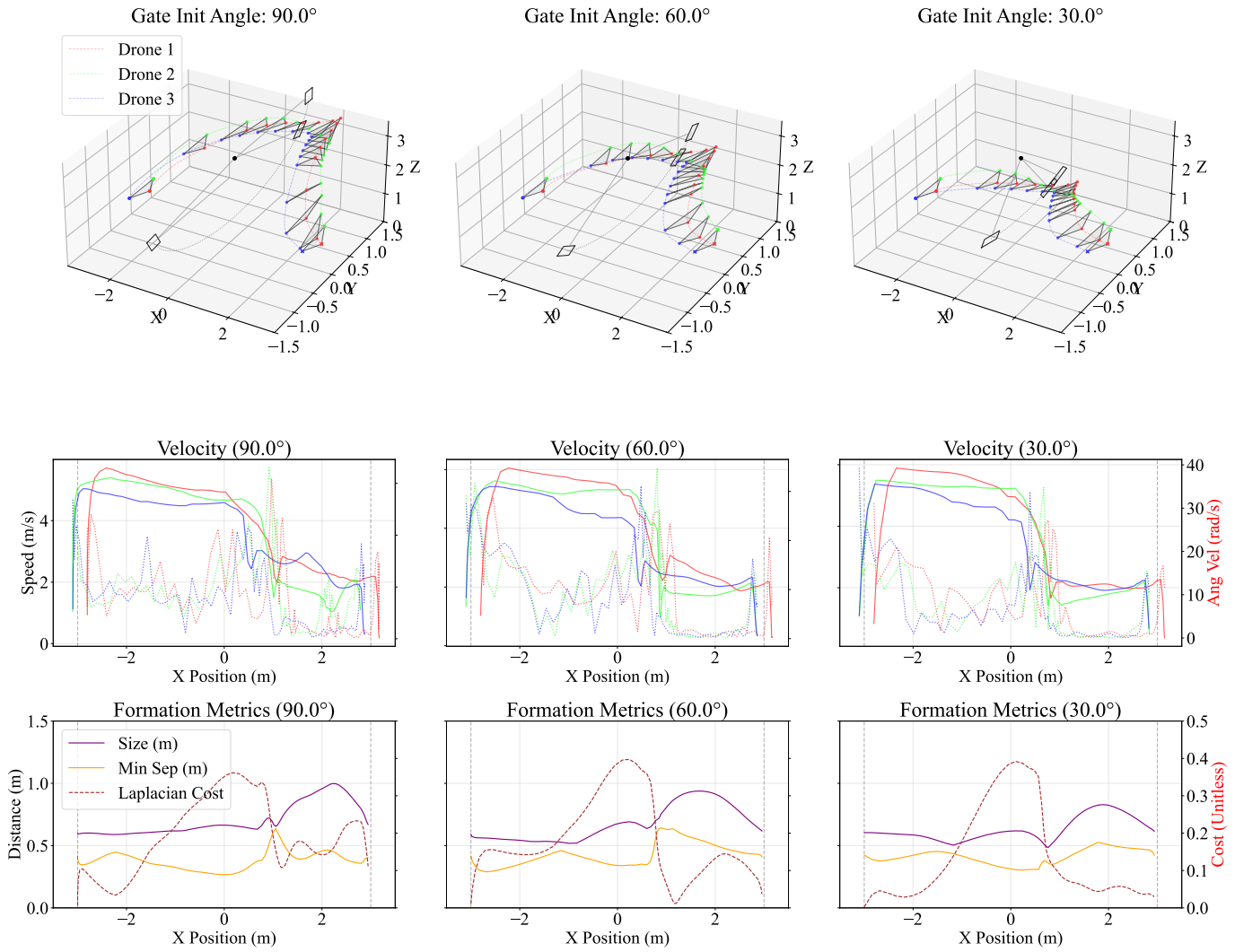



Fig. 3: Spatio-temporal visualization of the emergent "elastic formation" strategy across varying gate dynamics. The top row depicts the 3D flight trajectories, illustrating the synchronized approach, compression, and recovery phases. The middle row quantifies the agility through linear and angular velocity profiles. The bottom row correlates the geometric adaptation with formation metrics, highlighting the interplay between formation size, collision safety (Min Sep), and topological deviation (Laplacian Cost). The shaded regions indicate the critical gate traversal window.

execution. In contrast, GART-MAPPO demonstrates a superior balance of agility and precision. It executes highly aggressive maneuvers—indicated by an average angular velocity of 7.24 rad/s (nearly 3× that of High-MPC)—while maintaining the lowest Laplacian Cost (0.1542). This empirical evidence confirms that the learned elastic strategy effectively decouples topological integrity from rigid geometry, allowing the swarm to undergo rapid, coordinated deformation without breaking the underlying graph structure.

To further probe the robustness of the learned policy, we evaluated performance across a spectrum of task difficulties parameterized by the gate's initial swing angle (15° to 90°). As shown in Fig. 2, increasing the angle translates to higher target velocities, demanding sharper temporal precision. The *Inference Time* elucidates the critical scalability advantage of our method. While GART-MAPPO maintains a constant, millisecond-level inference ($\approx 1.5$ ms) regardless of difficulty, High-MPC exhibits a concerning linear growth, reaching nearly 100 ms at 90°. This indicates that as the dynamic constraints become more severe, the numerical solver requires significantly more iterations to converge, exacerbating the latency problem precisely when rapid responsiveness is most critical. Furthermore, the *Average Linear Speed* reveals an interesting behavioral trend: High-MPC attempts to match the increasing difficulty by accelerating, linearly increasing its speed from $\approx 0.8$ m/s to $\approx 3.3$ m/s to match GART-MAPPO at 90°. However, this aggression is futile due to the aforementioned computational lag. GART-MAPPO, conversely, maintains a consistently high speed ($\approx 3.0$ m/s) and, notably, a significantly higher average angular speed ( $\approx 7.5$ rad/s vs. High-MPC's $< 2$ rad/s). This sustained high angular velocity proves that GART-MAPPO solves the problem not by being

conservative, but by actively and continuously reorienting the formation topology to flow through the bottleneck, a capability that rigid baselines fundamentally lack.

### C. Behavior Analysis: Elastic Formation in Action

To elucidate the internal mechanism of the learned policy, Fig. 3 visualizes the spatio-temporal evolution of the swarm state during the critical traversal phase. The trajectories reveal a distinct "squeeze-and-recover" behavioral pattern that remains consistent across varying dynamic severities, from the moderate $30°$ swing to the extreme $90°$ scenario.

As the swarm approaches the critical interface (approximately $X = 0$ m), the policy autonomously modulates the formation geometry to satisfy the spatial bottleneck. This is quantitatively evidenced by the *Formation Size* metric, which exhibits a sharp, synchronized contraction from the nominal $0.6$ m to a compressed state of approximately $0.4$ m precisely at the point of traversal. Importantly, this geometric compression is not a passive deformation but an active, aggressive maneuver, as corroborated by the coincident spikes in angular velocity shown in the middle row. These spikes indicate that the agents are executing rapid roll and pitch adjustments to maintain flight stability while tightly clustering.

This maneuver fundamentally reflects the policy's ability to dynamically trade off conflicting objectives. The *Laplacian Cost*—representing the penalty for deviation from the ideal rigid structure—incurs a transient spike during the compression phase. However, this temporary violation of formation rigidity is strategically accepted by the agent to ensure task completion. Crucially, despite the extreme proximity required for traversal, the *Minimum Separation* is strictly maintained above the safety threshold (approx. $0.3$ m) throughout the maneuver. Immediately post-traversal, the formation size and Laplacian cost rapidly converge to their baseline values, demonstrating the system's elastic capability to recover its nominal topology after the environmental constraint is removed.

### D. Ablation Studies: Module Contribution Analysis

To rigorously dissect the functional contributions of the proposed architectural components, we performed a component-wise ablation study. The full GART-MAPPO framework was benchmarked against four variants to isolate the roles of spatial perception, temporal prediction, and task-context awareness.

The consolidated learning dynamics demonstrate the critical synergy of these modules. As illustrated in Fig. 4, GART-MAPPO achieves robust convergence and superior asymptotic rewards. In stark contrast, the MLP-MAPPO baseline suffers from a catastrophic training collapse at approximately 150M frames. This premature termination confirms that simple perceptrons lack the inductive bias required to stabilize policy updates in long-horizon, partially observable multi-agent interactions.

Specific module contributions are revealed through the breakdown analysis in Fig. 5:

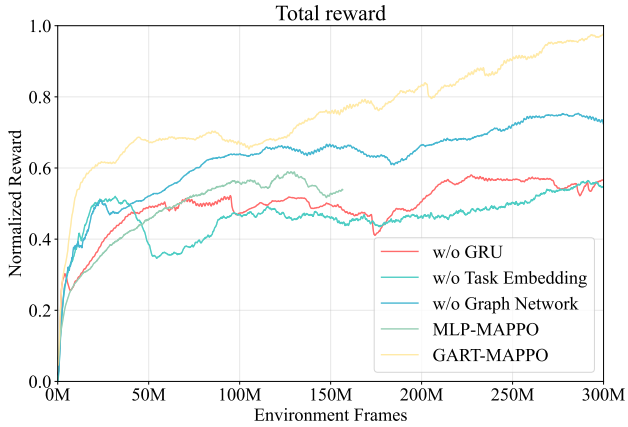


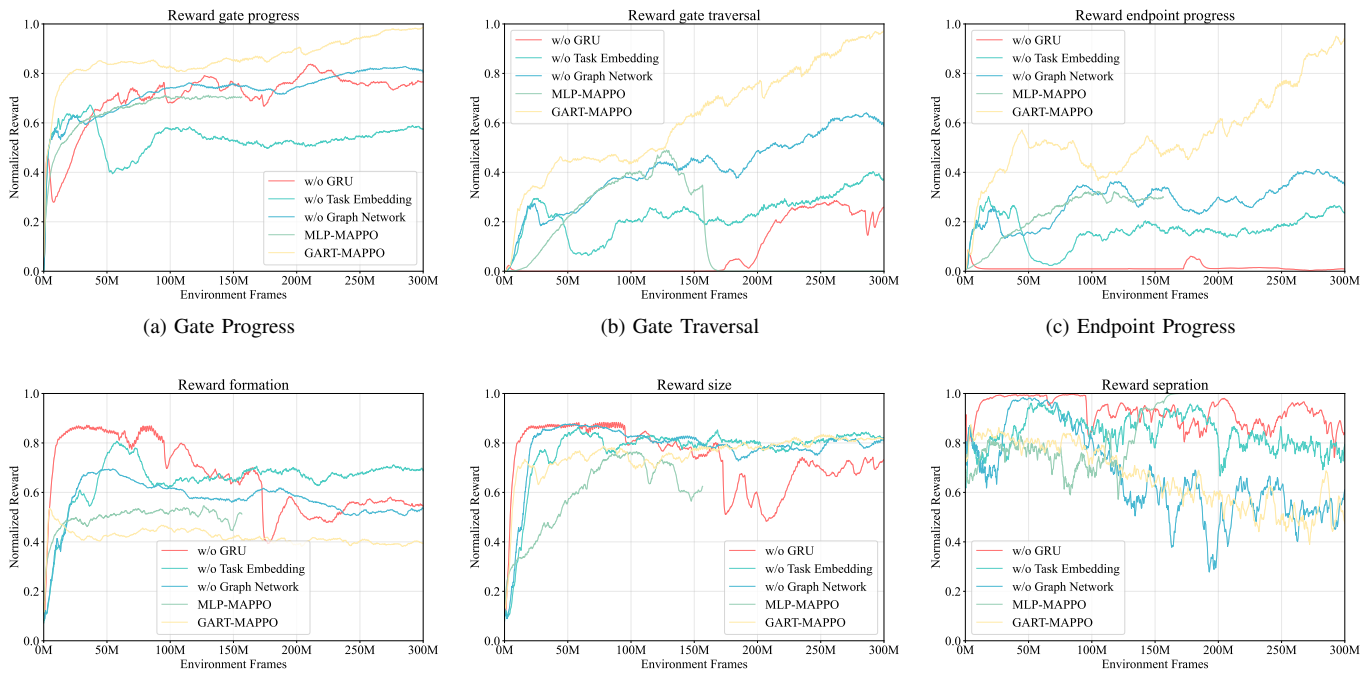

Fig. 4: Training Stability and Convergence Analysis. The proposed GART-MAPPO demonstrates superior asymptotic performance. A critical observation is the catastrophic collapse of the MLP-MAPPO baseline at approximately 150M frames, highlighting the necessity of the proposed architecture for maintaining long-horizon stability.

*1) Temporal Prediction and the "Approach-Traverse" Gap:* The ablation of the recurrent module (w/o GRU) provides compelling evidence for the necessity of temporal belief modeling. The top row of Fig. 5 reveals a critical discrepancy: while this variant performs competently in *Gate Progress* (Fig. 5(a)), indicating it can spatially locate and approach the gate, its performance flatlines near zero for *Gate Traversal* (Fig. 5(b)). This "capable of approaching but incapable of passing" behavior precisely isolates the failure mechanism: without a recurrent state to track the gate's latent oscillation phase, agents inevitably mistime their entry. Consequently, without successful traversal, the mission stalls, leading to zero *Endpoint Progress* (Fig. 5(c)).

*2) Task Awareness as a Soft-Switching Controller:* The w/o Task Embedding variant illustrates the risk of objective ambiguity. Despite maintaining high scores in spatial metrics like *Formation Reward* (Fig. 5(d)), it lags significantly in task execution metrics (Fig. 5(b)). This suggests that without explicit task-context attention to dynamically up-weight the traversal objective, the policy gravitates towards a conservative local optimum—prioritizing the "safe" reward of formation rigidity over the high-risk maneuver of dynamic traversal.

*3) Spatial Coherence via Graph Attention:* Finally, the w/o Graph Network variant exhibits a visible degradation in geometric topology. As shown in the bottom row of Fig. 5 (specifically *Formation Reward* and *Formation Size* in Figs. 5(d) and (e)), this variant displays higher variance and lower asymptotic stability. This confirms that ego-centric graph attention is essential for processing non-uniform neighbor dependencies, ensuring that the swarm remains cohesive even under the aggressive accelerations required for gate traversal.

Collectively, these results validate the "Perceive, Predict, and Cooperate" paradigm: spatial attention maintains topology, temporal recurrence enables dynamic timing, and task




(a) Gate Progress (b) Gate Traversal (c) Endpoint Progress

(d) Formation Reward (e) Formation Size (f) Min Separation

Fig. 5: Component-wise performance breakdown. Top Row (Task Dynamics): Comparing (a) *Gate Progress* vs. (b) *Gate Traversal* reveals that *w/o GRU* (Red) can approach the target but fails to predict the traversal window. Bottom Row (Spatial Topology): The *w/o Graph Network* variant (Blue) exhibits significant degradation and high variance in geometric metrics (d, e), confirming the role of attention in formation maintenance.

attention drives strategic objective switching.

## VI. Conclusion

In this paper, we presented GART-MAPPO, an end-to-end reinforcement learning framework for multi-agent coordination in dynamic, partially observable environments. The framework addresses the fundamental "Perceive, Predict, and Cooperate" challenge through three integrated modules: graph attention networks for spatial perception, recurrent neural networks for temporal prediction, and task-context attention for objective prioritization. By instantiating the framework on a challenging benchmark—formation flight through oscillating narrow gates—we demonstrated its capability to autonomously discover complex cooperative strategies. The learned "elastic formation" behavior, where agents dynamically compress to traverse narrow passages and recover geometry post-traversal, exemplifies the framework's ability to balance conflicting objectives without manual curriculum design. Extensive experiments validate that GART-MAPPO significantly outperforms state-of-the-art optimization-based and learning-based baselines, achieving near-perfect success rates with superior computational efficiency and robust generalization across varying difficulty levels. The ablation studies confirm the necessity of each architectural component. The generality of the proposed framework suggests broad applicability beyond the specific validation task. Future work will focus on transferring the learned policies to physical hardware through sim-to-real techniques, scaling the approach to larger agent groups operating in more cluttered environments, and extending the framework to heterogeneous multi-agent systems where agents possess diverse capabilities and pursue varied objectives.

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